

CASE REPORT

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Conservative management of prolonged air leak following tension pneumothorax in cavitary pulmonary tuberculosis: a case report

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Abstract

Background Secondary spontaneous pneumothorax is a recognized complication of pulmonary tuberculosis; however, presentation with tension physiology followed by prolonged air leak may create diagnostic and management challenges, particularly when tuberculosis is not known at presentation.

Case presentation We report a 20-year-old male with a 3 pack-year smoking history and no known prior lung disease who presented with dyspnea, left-sided pleuritic chest pain, productive cough, and constitutional symptoms. During the initial emergency assessment, unilateral chest findings raised concern for a left-sided pneumothorax. Urgent chest radiography obtained during this process showed a large left-sided pneumothorax with mediastinal shift, and emergency needle decompression was performed as a temporizing maneuver, followed directly by placement of a 28-Fr left chest tube. Persistent incomplete lung re-expansion and continuous air bubbling prompted further evaluation. Chest computed tomography revealed multiple left-sided cavitary lesions with surrounding parenchymal infiltrates, and sputum Ziehl–Neelsen staining was positive for acid-fast bacilli in three samples, supporting active cavitary pulmonary tuberculosis in the clinical and radiographic context. Standard anti-tuberculosis therapy was initiated. Despite early symptomatic improvement and substantial lung re-expansion, air leakage persisted. Because the patient maintained adequate oxygenation, controlled drainage, and no evidence of pleural sepsis or respiratory deterioration, conservative management was continued under thoracic surgical supervision with ambulatory drainage and close follow-up. The air leak resolved 61 days after hospital discharge, and the chest tube was removed without subsequent clinical recurrence.

Conclusions This case highlights that emergency decompression should not preclude continued etiologic evaluation when pneumothorax is accompanied by constitutional symptoms, incomplete lung re-expansion, persistent air leakage, or parenchymal abnormalities. It also illustrates that, in carefully selected clinically stable

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patients with adequate drainage and close supervision, prolonged air leak in active cavitary tuberculosis may be interpreted within the broader clinical trajectory rather than by duration alone.

Keywords Pulmonary tuberculosis, Cavitary tuberculosis, Tension pneumothorax, Secondary spontaneous pneumothorax, Persistent air leak, Prolonged air leak, Chest tube drainage, Case report

Background

Tension pneumothorax is a life-threatening presentation of pneumothorax that requires prompt recognition and urgent decompression [1]. In young patients without known lung disease, an apparently spontaneous pneumothorax may initially be presumed to represent a primary event, particularly when underlying parenchymal disease is not clinically evident at presentation [1]. However, the distinction between primary and secondary spontaneous pneumothorax is based on the absence or presence of clinically apparent lung disease, and patients classified as having primary spontaneous pneumothorax may still harbor unrecognized underlying lung abnormalities [2]. Pulmonary tuberculosis can underlie secondary spontaneous pneumothorax and, in some cases, progress to tension pneumothorax [3]. This distinction is clinically important because primary and secondary spontaneous pneumothoraces differ in morbidity, mortality, rates of hypoxia at presentation, and recommended management [2].

Pneumothorax and persistent pleural air leak are recognized complications in the setting of pulmonary tuberculosis [4, 5]. However, the published literature has more often described these complications in separate or broader clinical contexts than as a single integrated presentation [3, 4]. Although persistent air leak is generally defined and managed within time-based escalation frameworks, available guidance also acknowledges limited evidence regarding the optimal timing of intervention and the need for individualized decision-making [1, 5]. We report a case of tension pneumothorax as the initial presentation of active cavitary pulmonary tuberculosis, complicated by prolonged persistent pleural air leak despite early clinical and radiographic improvement under conservative management.

Case presentation

A 20-year-old male active smoker (3 pack-years), with no known prior lung disease, presented with a two-day history of progressive dyspnea and left-sided pleuritic chest pain associated with productive cough. He also reported recent fever, weight loss, and night sweats, without a history of chronic cough or known tuberculosis contact. On admission, he was tachycardic (130 bpm), tachypneic (35 breaths/min), and hypotensive (90/50 mmHg), with an oxygen saturation of 93% on room air. Physical examination revealed marked respiratory distress with intercostal retractions. Breath sounds were markedly diminished

over the left hemithorax, with absent tactile fremitus and hyperresonance to percussion, while the right lung examination was unremarkable. No jugular venous distension was observed.

During the first emergency assessment, the combination of dyspnea and unilateral chest findings raised concern for a left-sided pneumothorax, although the diagnosis was not clinically definitive at that point. The thoracic surgery team was called immediately, and an urgent chest radiograph was obtained during this process. The radiograph demonstrated a large left-sided pneumothorax with mediastinal shift (Fig. 1), consistent with tension pneumothorax. Emergency needle decompression was then performed as a temporizing maneuver to relieve tension physiology while tube thoracostomy was being prepared, followed directly by placement of a 28-Fr left chest tube connected to an underwater seal drainage system. Laboratory investigations obtained following initial stabilization demonstrated leukocytosis ($12,800/\text{mm}^3$) with neutrophilia (89%) and elevated C-reactive protein (138 mg/L), while hemoglobin, renal function, liver enzymes, and electrolytes were within reference ranges. Despite initial intervention, post-procedural imaging demonstrated incomplete lung re-expansion with residual pneumothorax, left-sided interstitial infiltrates, and subcutaneous emphysema (Fig. 2), and continuous air bubbling persisted in the drainage system.

On the day after chest tube placement, and in view of the incomplete re-expansion, associated constitutional symptoms, and radiographic parenchymal abnormalities, chest computed tomography was performed, revealing a cavitary lesion at the apex of the left upper lobe and multiple cavitary lesions in the left lower lobe, with surrounding parenchymal infiltrates and mild left pleural effusion (Fig. 3), while the right lung remained unremarkable. Sputum samples were obtained for microbiological analysis. Empirical intravenous ceftriaxone and levofloxacin were initiated for presumed severe bacterial pleuropulmonary infection while microbiological testing, including acid-fast bacilli staining, was pending. Approximately three days after CT imaging, microbiological evaluation of sputum samples demonstrated acid-fast bacilli on three Ziehl–Neelsen stains, supporting the diagnosis of active pulmonary tuberculosis in the clinical and radiographic context. GeneXpert MTB/RIF testing, mycobacterial culture, and drug-susceptibility testing were not performed. Standard anti-tuberculosis therapy was initiated, consisting of isoniazid, rifampicin,

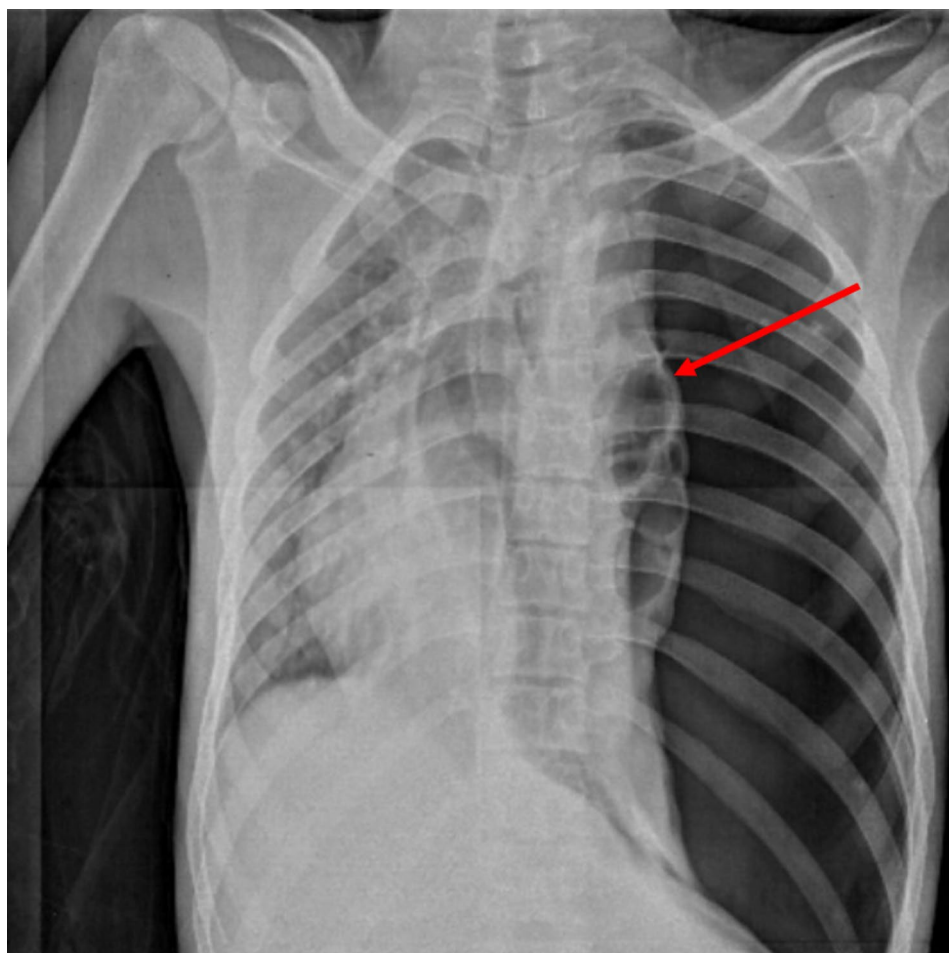


Fig. 1 Initial chest radiograph showing a large left-sided pneumothorax with contralateral mediastinal shift, consistent with tension pneumothorax. The red arrow indicates the collapsed left lung margin

pyrazinamide, and ethambutol for two months, followed by isoniazid and rifampicin to complete the standard treatment course.

Within one week of chest tube placement, follow-up radiography demonstrated substantial re-expansion of the left lung (Fig. 4), accompanied by marked clinical improvement, resolution of dyspnea, and improvement in oxygenation. Despite this early improvement, air leakage persisted, with continuous bubbling observed in the chest drainage system during hospitalization. Air-leak assessment was clinical, without digital quantification or standardized grading.

Because the patient remained clinically stable after the initial improvement, conservative management was continued under thoracic surgical supervision. Further surgical or bronchoscopic escalation was considered as a potential option if the clinical course became unfavorable; however, it was deferred because the patient maintained adequate oxygenation on room air, had substantial radiographic re-expansion, and showed controlled drainage without pleural sepsis, recurrent respiratory

compromise, or other urgent indications for intervention. Given this favorable early trajectory and the ability to maintain controlled drainage outside the hospital, he was discharged with ambulatory chest drainage using a portable one-way drainage bag system.

During the outpatient drainage period, clinical follow-up was maintained approximately two to three times weekly to assess clinical status, drainage function, and air-leak persistence. During follow-up, the air leak gradually decreased, became intermittent, and eventually ceased. At 61 days after hospital discharge, bubbling had been absent for 24 h, the air leak was considered clinically resolved, and the chest tube was removed.

After chest tube removal, clinical follow-up was maintained twice during the first week and then weekly for approximately three weeks. The patient remained clinically well without subsequent clinical recurrence and continued anti-tuberculosis therapy to complete the standard treatment course.

The key diagnostic and management events are summarized in Table 1.



Fig. 2 Post-procedural chest radiograph showing an in-situ left chest tube, residual left-sided pneumothorax with incomplete lung re-expansion, persistent left-sided parenchymal opacities, and subcutaneous emphysema. Red arrows indicate the residual pleural air space and the chest tube

Discussion

Although pulmonary tuberculosis is a recognized cause of secondary spontaneous pneumothorax, this remains an uncommon complication of active disease [4, 6]. Presentation as tension pneumothorax has also been described, although it appears to be reported less frequently [3]. Persistent air leak related to pleural or bronchopleural communication has likewise been reported in tuberculous pleuropulmonary disease [7]. The clinical significance of the present case lies less in any isolated component than in the diagnostic and management challenge created by their sequence. A patient without

known tuberculosis initially presented with tension pneumothorax, and the subsequent persistence of air leakage, together with incomplete lung re-expansion and parenchymal abnormalities, prompted continued etiologic evaluation, which supported the diagnosis of active cavitary pulmonary tuberculosis in the appropriate clinical and radiographic context.

In a young patient without known prior lung disease, the initial presentation could have been interpreted as an apparently spontaneous pneumothorax before the subsequent course clarified the underlying disease process [4]. Pleural decompression addressed the immediate

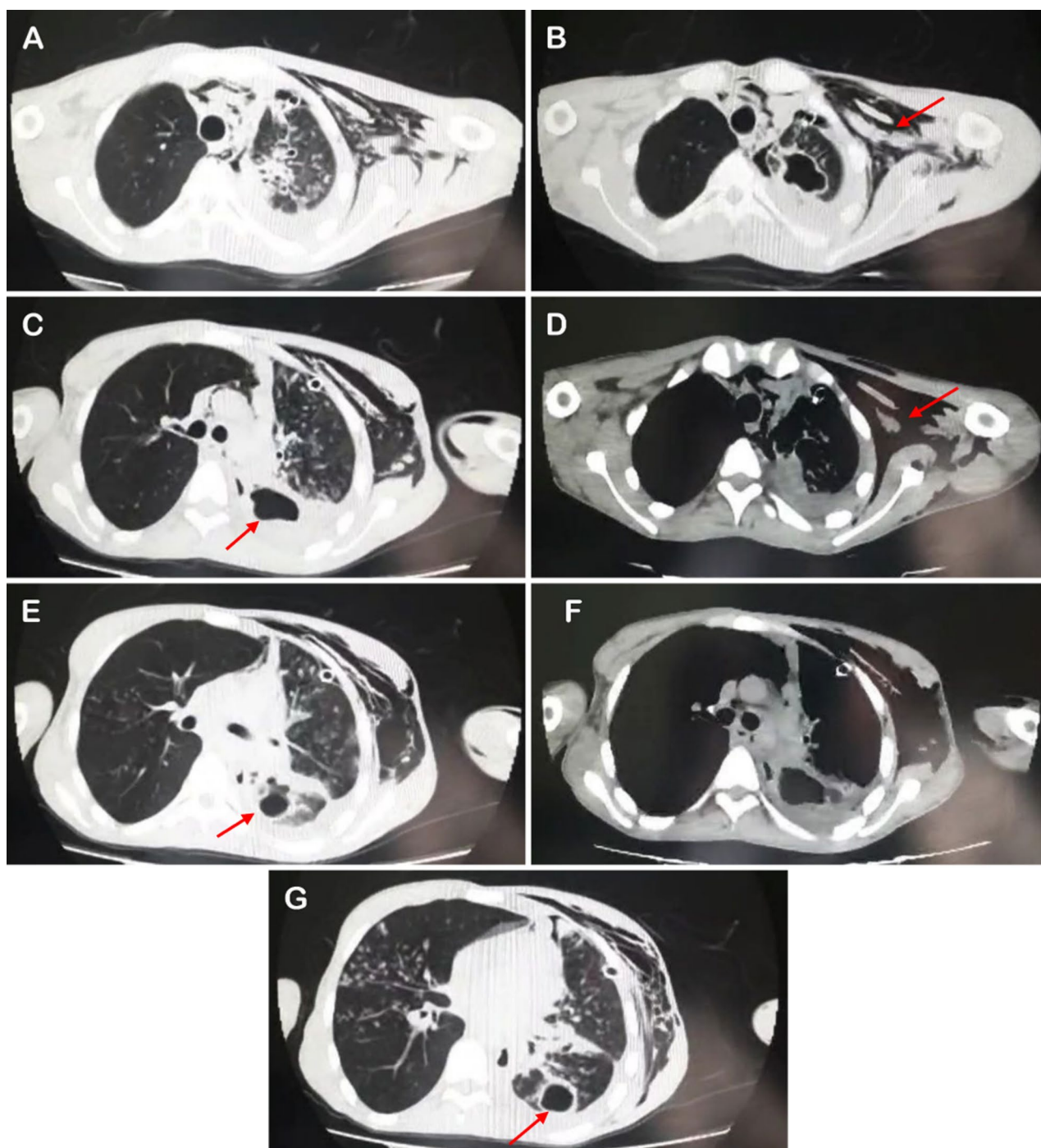


Fig. 3 Chest computed tomography showing left-sided cavitory lesions with surrounding parenchymal infiltrates, mild pleural effusion, subcutaneous emphysema, and an in-situ left chest tube. Red arrows indicate representative cavitory lesions and subcutaneous emphysema. Panels A–G show axial sections at different thoracic levels

threat, but it did not fully explain the broader clinical picture. Constitutional symptoms, incomplete lung re-expansion, persistent air leakage, and associated parenchymal abnormalities argued against viewing the episode as an isolated pleural event. Chest computed tomography therefore played a central role in redirecting the

diagnostic evaluation toward active cavitory pulmonary tuberculosis. The microbiological findings supported this diagnosis in the clinical and radiographic context. Accordingly, pleural decompression should not terminate etiologic evaluation when the subsequent course suggests an underlying parenchymal disease process [4, 8].



Fig. 4 Follow-up chest radiograph showing substantial re-expansion of the left lung with an in-situ chest tube, persistent left-sided parenchymal opacities, and residual subcutaneous emphysema

Persistent air leak is generally defined as continued bubbling after 3–5 days of chest drainage and remains a recognized challenge in spontaneous pneumothorax [9]. However, such time-based definitions do not by themselves determine how persistent air leak should be interpreted or managed, particularly in complex disease-specific settings where evidence remains limited [9, 10]. In active cavitary tuberculosis, ongoing parenchymal necrosis, communication with the airways, and pleural extension may all contribute to delayed closure of the air leak [11]. In the present case, the possibility of bronchopleural or alveolopleural communication was considered clinically because of persistent bubbling and cavitary disease. However, no direct fistulous tract was demonstrated on CT, and bronchoscopy was not performed; therefore, the finding was described as persistent air leak rather than confirmed bronchopleural fistula. Accordingly, ongoing bubbling in this setting should not be interpreted in isolation as evidence of treatment failure [6].

Contemporary management frameworks generally support early escalation of persistent air leak, particularly when the patient is fit for definitive intervention [9]. Yet this approach cannot be applied mechanically when prolonged air leakage occurs alongside otherwise favorable clinical recovery [9, 12]. In our patient, continued conservative management was supported by rapid symptomatic improvement, adequate oxygenation on room air, substantial radiographic re-expansion of the affected lung, and controlled ongoing drainage. There was no recurrent

Table 1 Timeline of key clinical events

Time point	Event
Day 0	Presentation to the emergency department with dyspnea and unilateral chest findings. Urgent chest radiography showed a large left-sided pneumothorax with mediastinal shift. Emergency needle decompression was performed, followed by placement of a 28-Fr left chest tube connected to an underwater seal drainage system.
Day 1	Chest CT was performed after chest tube placement and showed left-sided cavitary lesions with surrounding parenchymal infiltrates, mild pleural effusion, subcutaneous emphysema, and an in-situ chest tube.
Day 4	Sputum Ziehl–Neelsen staining was positive for acid-fast bacilli, and standard anti-tuberculosis therapy was initiated.
Within the first week	Clinical improvement and substantial radiographic re-expansion of the left lung were observed, although air leakage persisted.
At hospital discharge	The patient remained clinically stable with adequate oxygenation and controlled drainage. He was discharged with ambulatory chest drainage using a portable one-way drainage bag system.
During outpatient drainage	Clinical follow-up was maintained approximately two to three times weekly to assess clinical status, drainage function, and air-leak persistence. The air leak gradually decreased and became intermittent.
61 days after hospital discharge	Bubbling had been absent for 24 h, the air leak was considered clinically resolved, and the chest tube was removed.
After chest tube removal	Clinical follow-up was maintained twice during the first week and then weekly for approximately three weeks. The patient remained clinically well without subsequent clinical recurrence.

Abbreviation: CT computed tomography

respiratory compromise, superimposed pleural infection, or other urgent indication for definitive intervention. This decision was made under thoracic surgical supervision, with surgical or bronchoscopic escalation reserved for an unfavorable clinical course. At the treating institution, thoracic surgical assessment and supervised chest-drainage follow-up were available, whereas advanced bronchoscopic air-leak interventions were not routinely available. The presence of active cavitary tuberculosis, with inflamed and potentially friable parenchyma, was also considered when weighing the risks and benefits of non-urgent intervention. The ability to continue supervised outpatient management further supported ongoing drainage and observation over immediate invasive management. In this context, leak duration alone was insufficient to define management failure or justify immediate definitive intervention [9, 13]. This should not be interpreted as a general recommendation to delay intervention, but as an example of individualized management in a clinically improving patient.

Previously reported series of tuberculosis-associated pneumothorax provide useful context for the

present case, although they do not fully mirror its clinical sequence. In the 53-patient series by Shamaei et al., chest-tube drainage alone relieved most surgically managed patients, with a median drainage duration of 13 days, whereas some patients required open drainage or more invasive procedures because of empyema, lung destruction, or bronchopleural fistula [7]. Freixinet et al. similarly reported a mean pleural drainage duration of 12.9 days, with surgical resection required in 28% of patients, mainly for persistent air leakage or recurrence [4]. In another series, Singh and Tiwari found that underwater-seal intercostal drainage was successful in most patients, with a mean drainage duration of 17.14 days, while 7.7% were referred for thoracic surgical management because of non-resolution of pneumothorax and persistent bronchopleural fistula [14]. Against this background, the present case is notable for the initial presentation as tension pneumothorax in previously undiagnosed cavitary tuberculosis, the substantially prolonged drainage course, and eventual resolution without surgical or bronchoscopic escalation. This variability, together with the absence of a clearly defined escalation threshold for persistent air leak in disease-specific settings [9, 10, 13], supports the educational value of reporting carefully monitored conservative courses when decisions are guided by clinical trajectory rather than duration alone.

Limitations

Tuberculosis was diagnosed on the basis of three positive sputum Ziehl–Neelsen smears in the appropriate clinical and radiographic context, without GeneXpert MTB/RIF testing, mycobacterial culture, or drug-susceptibility testing; therefore, molecular confirmation and drug-resistance assessment were not available. This limitation should be considered when interpreting diagnostic certainty and may reduce the generalizability of the case to settings where molecular confirmation, culture, and drug-susceptibility testing are routinely available. Air-leak assessment was qualitative and based on clinical observation of bubbling in the drainage system rather than digital quantification or standardized grading. Although persistent bubbling and cavitary disease raised concern for bronchopleural or alveolopleural communication, no direct fistulous tract was demonstrated on CT and bronchoscopy was not performed. Follow-up after chest tube removal was clinical, without a follow-up chest radiograph. As a single case, these limitations preclude defining a general threshold for escalation or a standard duration for conservative management. The case should therefore be interpreted as an example of individualized management in a clinically stable patient rather than evidence that prolonged air leak duration alone can be safely disregarded.

Conclusion

This case highlights that previously undiagnosed active cavitary pulmonary tuberculosis may initially present as tension pneumothorax and may be complicated by prolonged persistent air leak. Pleural decompression addresses the immediate emergency, but persistent air leakage, incomplete lung re-expansion, constitutional symptoms, or parenchymal abnormalities should prompt continued etiologic evaluation. In carefully selected patients who remain clinically stable with adequate oxygenation, controlled drainage, lung re-expansion, and no evidence of pleural sepsis or respiratory deterioration, prolonged air leak may be interpreted within the broader clinical course rather than by duration alone. This case supports individualized, closely supervised decision-making rather than a general recommendation to delay surgical or bronchoscopic intervention.

Abbreviations

CT	Computed tomography
TB	Tuberculosis

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Not applicable.

Key Clinical Message

Tension pneumothorax may be the initial presentation of previously undiagnosed cavitary pulmonary tuberculosis; therefore, emergency decompression should not preclude continued etiologic evaluation when the subsequent course is atypical. In carefully selected clinically stable patients, persistent air leak after decompression should be interpreted within the broader clinical trajectory, including lung re-expansion, drainage control, absence of pleural sepsis, and respiratory stability, rather than by duration alone.

Authors' contributions

AA conceived the case report, collected and interpreted the clinical data, performed the literature review, drafted the manuscript, and led the final revision. MAI assisted in data collection and manuscript drafting, coordinated communication with the supervisor, and revised the manuscript. LAI contributed to clinical data interpretation and reviewed the manuscript for clinical accuracy. MB supported data organization and critically revised the manuscript. MAS contributed to case follow-up and manuscript review. MSh contributed to data acquisition and interpretation and participated in manuscript drafting and critical revision. BD provided overall supervision, critical intellectual input, and approved the final version of the manuscript. All authors read and approved the final manuscript and agree to be accountable for all aspects of the work.

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Data availability

The data supporting the findings of this case report are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Written informed consent was obtained from the patient for publication of this case report and any accompanying images. A copy of the written consent is available for review by the Editor-in-Chief of this journal on request.

Competing interests

The authors declare no competing interests.

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